

	PhET Greenhouse Effect

•

Connor Petri

1. Hypothesize why the inside of a car feels so much warmer than its surroundings on sunny days.

Light entering the car is trapped by non-transparent surfaces, which radiate that heat into the air. The car is closed, so the air cannot circulate. This means the same air is absorbing the heat the whole time the car is parked.

2. Go to <http://phet.colorado.edu/en/simulation/greenhouse>. Click “Run Now”

Part I: A Greenhouse Simulation

1. Select the “Glass Layers” tab.
2. What do the yellow stars represent?

Answer: Radiation from the Sun

3. What do the red stars represent?

Answer: Infrared radiation

4. Both the yellow and red stars represent forms of energy in the form of photons: the yellow are visible light, the red are heat.

Technically infrared radiation is not heat, rather one of many ways heat can be transferred.

5. Record the approximate temperature “inside the greenhouse” before adding glass panes.

Answer: -19 °C

6. Add one glass pane. What do the sunlight photons do when they hit the glass from the top?
Pass through unchanged if we ignore refraction.

7. What do the infrared photons do when they hit the glass from the bottom? Be specific.

Some of them pass through, but many are reflected back into the greenhouse.

8. What is the new temperature “inside the greenhouse?”

Answer: 32 °C

9. Based on the observations of the photons, why does the temperature go up so much?
The infrared radiation is trapped within the greenhouse, which transfers thermal energy to the air and ground within.

10. What happens to the temperature as additional glass panes are added?

Answer: The temperature rises.

11. Explain why this happens by observing the photons.
With each glass pane added, less and less radiation can escape. This drives up temperature within the greenhouse.

12. Before proceeding to the earth, predict how what you have discovered regarding greenhouses might apply to the earth and its atmosphere.

Since infrared radiation is a method of thermal energy transfer, it won't just heat up the ground, it heats up anything it touches, including the atmosphere. The first law of thermodynamics states that energy cannot be created or destroyed in a closed system, meaning if less infrared radiation can escape the atmosphere, then it gets absorbed by it and turned into heat.

Part II: The Earth Simulation

1. Select the “Greenhouse Effect” tab. Which greenhouse gases are considered by the simulation?

Answer: Water, Carbon Dioxide, Methane, and Nitrous Oxide

2. Which time period do the default conditions represent?

Answer: The Present

3. The thermometer represents the average global temperature. What is the average global temperature for the “today” simulation?

Answer: 15 °C

4. Is the behavior of the photons more similar to the greenhouse simulation with or without glass panes?

Answer: It resembles the simulation with the glass panes. Some of the IR radiation is not escaping into space.

5. Reduce the greenhouse gas concentration to “None”. Is the behavior of the photons more similar to the greenhouse simulation with or without glass panes?

Answer: Without

6. What is the average global temperature?

Answer: -18 °C

7. Considering the behavior of the photons, why does the temperature drop so much?

The atmosphere allows 100% of the IR hitting it to escape into space, therefore the atmosphere never heats because it isn't absorbing any light.

8. Increase the greenhouse gas concentration to “Lots.” What is the average global temperature?

Answer:

9. Considering the behavior of the photons, why does the temperature increase?

Most of the IR is being absorbed by the atmosphere, which heats it up.

Part III: Incremental Analysis. Write a paragraph that answers the following questions:

1. How does the “greenhouse effect” affect temperature on the earth?

As the concentration of greenhouse gases in the atmosphere increases, the less “transparent” it is to infrared wavelengths. The more IR the atmosphere absorbs, the more thermal energy is transferred to it. The increase in global temperatures continues to shrink Earth's ice caps. This reduces the average surface albedo of the Earth, causing the surface to absorb more light. This in turn increases the amount of IR the Earth emits into the atmosphere, which further contributes to atmospheric temperature increases.

2. How is the “greenhouse effect” similar to blankets on a bed?

A blanket keeps you warm by reducing the amount of heat lost to the environment. Your body radiates heat into the blanket and the air inside it, and the blanket acts as a thermal insulator. Traditional blankets absorb heat, which reduces thermal energy transfer to the air around the user. There are also survival blankets, which reflect the IR radiation back at the user, which also prevents thermal energy loss. The Earth's atmosphere is like it's blanket. By absorbing or reflecting IR, the atmosphere prevents the loss of thermal energy into space.

3. Is the “greenhouse effect” good or bad for the earth?

The Earth is a rock that is not alive, therefore defining what is good or bad for it can be subjective. The greenhouse effect has always been present, however a runaway greenhouse effect can be deadly for the organisms that call Earth home. However, if humans were to go extinct or stop producing greenhouse gases/destroying carbon absorbing organisms, the planet would eventually recover after an immense period of time.

Part IV: Analyze how Greenhouse Gases affect Earth's temperature and identify processes that might increase or decrease Earth's temperature

Ice Age

1. Click on the "Ice Age" button and record the minimum temperature: 18 °C
2. Record: CO₂: 180 ppm CH₄: .380 ppm N₂O: .215 ppm
3. What happens to the sunlight photons as they hit the ground? Are all of the sunlight photons being reflected back into space?

The ice is reflecting the photons back into space rather than absorbing their energy as heat and radiating IR into the atmosphere.

4. What happens to the temperature when you add clouds?

The temperature increases the more clouds there are.

5. How are the photons affected by adding clouds?

Some of the visible light is reflected back into space, but far more IR is trapped under the clouds. This drives the temperature up.

1750

1. Click on the "1750" button and record the minimum temperature: 11 °C
2. Record: CO₂: 280 ppm CH₄: .730 ppm N₂O: .270 ppm
3. What happens to the sunlight photons as they hit the ground?

More visible light is absorbed than is reflected.

4. What happens to the temperature when you add clouds?

The temperature increases slightly, but not as much as during the ice age.

5. How are the photons affected by adding clouds?

The clouds absorb and reflect IR, however the atmosphere does too. The higher concentration of greenhouse gases means the difference between clouds and no clouds is less dramatic.

Today

1. Click on the "Today" button and record the minimum temperature. 15 °C
2. Record: CO₂: 388 ppm CH₄: 1.843 ppm N₂O: 0.317 ppm
3. What happens to the sunlight photons as they hit the ground?

They are absorbed almost completely.

4. What happens to the temperature when you add clouds?

It cools down slightly.

5. How are the photons affected by adding clouds?

There is little difference in the amount of IR that escapes, but the clouds absorb some of the visible light, reducing temperatures beneath them.

Comparison

1. Describe how, according to the model, the effects of Greenhouse Gases on the Earth's atmosphere have changed from the Ice Age to today.

The concentration of CO₂ has increased by a factor of 1.88, the CH₄ concentration has increased by a factor of 4.85, and the N₂O concentration has increased by a factor of 1.47

2. What assumptions do you think the model is making? Do you think those assumptions are valid?

The model assumes that there are only 4 greenhouse gases, that the ground doesn't reflect any light unless it's ice, and that IR and heat are the same thing. This is very good for making a simple simulation for the purposes of education, but would be wildly inaccurate if one were to try to use it to model reality.

3. What would happen if there were no Greenhouse gases? (Adjust the Greenhouse Gas Concentration Level to None.)

All of the IR would escape into space without being absorbed into the atmosphere. This would drive atmospheric temperatures down.

4. What will happen if Greenhouse gases increase in the future? (Adjust the Greenhouse Gas Concentration Level to Lots.)

Very little IR would escape the atmosphere, which would drive atmospheric temperatures up.

5. Identify some natural or human processes that might increase or decrease the amount of greenhouse gases in the atmosphere.

Factory farming, combusting gasoline, burning coal, killing plant life, anything that consumes a large amount of electricity not generated from a renewable source, and unregulated capitalism.

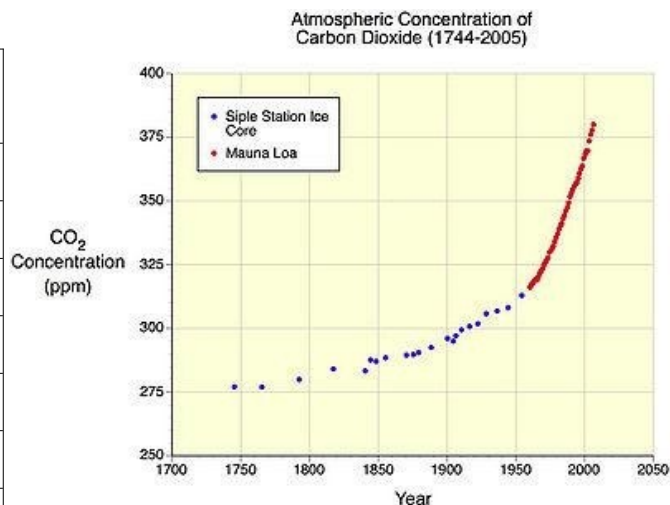
Part V: Scientific Question: Which atmospheric gas (CH₄, CO₂, H₂O, N₂, or O₂) is the best absorber of infrared photons?

Lab Background Information

Table 1: Average composition of the Earth's atmosphere up to an altitude of 25 km.

Gas Name	Chemical Formula	Percent Volume
Nitrogen	N ₂	78.08%
Oxygen	O ₂	20.95%
*Water	H ₂ O	0 to 4%
Argon	Ar	0.93%
*Carbon dioxide	CO ₂	0.0360%
Neon	Ne	0.0018%
Helium	He	0.0005%
*Methane	CH ₄	0.00017%
Hydrogen	H ₂	0.00005%
*Nitrous oxide	N ₂ O	0.00003%
*Ozone	O ₃	0.000004%
* variable gases		

- **CH₄** is a very strong greenhouse gas. Since 1750, methane concentrations in the atmosphere have increased by more than 150%. The primary sources for the additional methane added to the atmosphere (in order of importance) are: rice cultivation; domestic grazing animals; termites; landfills; coal mining; and, oil and gas extraction.
- The volume of **CO₂** has increased by over 35% in the last three hundred years. This increase is primarily due to human activities such as combustion of fossil fuels, deforestation, and other forms of land-use change. It is now fact—the increase is causing global warming through an enhancement of the greenhouse effect.
- **H₂O** (water vapor) varies in concentration in the atmosphere both spatially and temporally. Water vapor has several very important functional roles on our planet. For example, the condensation of water vapor creates precipitation that falls to the Earth's surface providing needed fresh water for plants and animals. Additionally, it helps warm the Earth's atmosphere through the [greenhouse effect](#).
- N₂ is removed from the atmosphere and deposited at the Earth's surface mainly by specialized nitrogen fixing bacteria, and by way of lightning through precipitation. The addition of this nitrogen to the Earth's surface soils and various water bodies' supplies much needed nutrition for plant growth.
- O₂ is exchanged between the atmosphere and life through the processes of photosynthesis and respiration. Photosynthesis produces oxygen when carbon dioxide and water are chemically converted into glucose with the help of sunlight.
- Remember, infrared radiation is heat.



Tools:

1. Click on the “Photon Absorption” Tab
2. Observe what happens when you move the slider on the photon emitter.
3. Observe what happens when you change which atmospheric gas you are examining.
4. After examining how this application works for a few minutes, you are now ready to begin your experiment!

Design and Execute An Experiment:

1. Develop a methodology for testing the different elements for photon absorption (infrared only).
Write the steps for your procedure below:

a.
b.
c.
d.
e.
f.

2. Summarize your results in the following table (disregard error and uncertainty, but ensure you are clear in what your measurements consist of).

	CH₄	CO₂	H₂O	N₂	O₂

3. Write a conclusion that answers the question, “Which atmospheric gas (CH₄, CO₂, H₂O, N₂, or O₂) is the best absorber of infrared photons?”

4. Based on your experiment, what you have learned, and the information provided above, what are the implications of this information?
